

SHIVAM PATEL

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EDUCATION

Carnegie Mellon University

2024-28 (Expected)

PhD in Electrical and Computer Engineering

Advisor: Prof. Gauri Joshi

Indian Institute of Technology Bombay

2020-24

Bachelor of Technology in Electrical Engineering

Advisor: Prof. Vivek Borkar; CPI: 9.66/10

RESEARCH INTERESTS

Memory and Compute Efficiency in LLMs, Hardware-Algorithm Co-Design for Inference Systems.

Previously: Optimization, Stochastic Systems, Game Theory.

WORK EXPERIENCE

AI Research Intern

Robert Bosch GmbH, Pittsburgh

May - August '26

Advisor: Bingqing Chen.

Graduate Researcher

Carnegie Mellon University, Pittsburgh

August '24 - Present

Advisor: Prof. Gauri Joshi. Working on compute and memory efficiency of LLMs as a PhD Candidate at CMU.

Visiting Undergraduate Researcher

USC Viterbi School of Engineering, Los Angeles

May - Aug '23

Advisor: Prof. Meisam Razaviyayn. Worked on statistical fairness and robust model objectives.

PUBLICATIONS AND PREPRINTS

- [6] Beyond Accuracy and Cost: Latency-Aware LLM Query Routing for Dynamic Workloads
Shivam Patel^{*}, Akaash Parthasarathy^{*}, Ankur Mallick, Gauri Joshi. Preprint, 2026.
- [5] LOCUS: Low-Dimensional Model Embeddings for Efficient Model Exploration, Comparison, and Selection
Shivam Patel, William Cocke, Gauri Joshi. AdaptFM-ICML 2026. [🔗](#)
- [4] Federate the Router: Learning Language Model Routers with Sparse and Decentralized Evaluations
Baris Askin^{*}, **Shivam Patel**^{*}, Anupam Nayak^{*}, Andrea Vigano, Jiin Woo, Gauri Joshi, Carlee Joe-Wong. arXiv preprint, 2026. [🔗](#)
- [3] ProxRouter: Proximity-Weighted LLM Query Routing for Improved Robustness to Outliers
Shivam Patel, Neharika Jali, Ankur Mallick, Gauri Joshi.
International Conference on Artificial Intelligence and Statistics, AISTATS 2026. [🔗](#)
- [2] An Asymptotic CVaR Measure of Risk for Markov Chains
Shivam Patel, Vivek S. Borkar.
IEEE Signal Processing and Communications Conf., SPCOM 2024. [🔗](#)
- [1] f -FERM: A Scalable Framework for Robust Fair Empirical Risk Minimization
Sina Baharlouei, **Shivam Patel**, Meisam Razaviyayn.
The Twelfth International Conference on Learning Representations, ICLR 2024. [🔗](#)

^{*} Equal Contribution.

RELEVANT COURSEWORK

At CMU: Algorithms for Language Modeling, Generative AI, Statistical Methods in Machine Learning,
Large Scale Optimization for Distributed ML

At IITB: Stochastic Optimization, Markov Chains and Queueing Systems, Advanced ML, Information Theory
Estimation and Identification, Reinforcement Learning, Game Theory

SCHOLASTIC ACHIEVEMENTS

- Exemplary PhD Qualifying Examination Performance by ECE, CMU *(Spring 26)*
- Awarded the William Messner Endowed Fellowship at ECE, CMU *(2025-26)*
- Conferred the Carnegie Institute of Technology Dean's Fellowship at CMU *(2024-25)*
- Awarded the IUSSTF-Viterbi Scholarship for pursuing summer research at USC Viterbi School
by Department of Science & Technology, Govt. of India. One of 15 selected students across India. *(2023)*
- Accorded the Kishore Vaigyanik Protsahan Yojana (KVPY) fellowship by IISc and DST India *(2020)*
- Achieved an All India Rank 219 in JEE Advanced among 0.225 million candidates *(2020)*